

Phase I and Phase II enzymes are responsible for breaking down medications and toxicants. Please see the linked articles for details and complete references.

Phase I Detoxification

[Article: CYP1A1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP1A1	rs1048943	TT	C	0.04	CYP1A1*2; higher risk of oral cancer & lung cancers in smokers; higher activity in estrogen metabolism
CYP1A1	rs1799814	GG	T	0.04	CYP1A1*4; altered enzyme activity
CYP1A1	rs2606345	AA	A	0.59	Decreased risk of testicular cancer
CYP1A1	rs2472297	CC	T	0.2	Associated with increased coffee consumption

[Article: CYP1A2](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP1A2	rs762551	CA	C	0.31	A/C: Intermediate caffeine metabolism; C/C: Slow caffeine metabolism
CYP1A2	rs12720461	CC	T	0.002	Decreased CYP1A2 enzyme activity
CYP1A2	rs72547517	--	A	0	Decreased CYP1A2 enzyme activity
CYP1A2	rs72547515	--	A	0.00004	Decreased CYP1A2 enzyme activity

[Article: CYP2A6](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2A6	rs1801272	AA	T	0.02	CYP2A6*2 reduced activity
CYP2A6	rs5031017	--	A	0.002	CYP2A6*5 non-functioning variant
CYP2A6	rs5031016	--	G	0.004	CYP2A6*7 non-functioning variant
CYP2A6	rs28399444	--	D	0	CYP2A6*20 non-functioning variant
POR	rs1057868	CC	T	0.28	May increase CYP2A6 enzyme activity in people with normal CYP2A6 variants
CYP2A6	rs28399453	--	A	0.05	Increased CYP2A6 activity

[Article: CYP2B6](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2B6	rs3745274	TT	T	0.25	Decreased CYP2B6 activity, may impact dosages needed for ketamine, methadone, efavirenz, and other drugs; decreased risk of fatty liver disease
CYP2B6	rs2279343	--	G	0.25	Reduced CYP2B6 activity, may impact dosages needed for ketamine, efavirenz, and other drugs
CYP2B6	rs28399499	TT	C	0.004	Significantly reduced enzyme activity
CYP2B6	rs3211371	--	T	0.09	Reduced enzyme activity, especially in women
CYP2B6	rs12721655	--	G	0.003	Reduced enzyme activity

[Article: CYP2C8](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2C8	rs10509681	TT	C	0.1	CYP2C8*3 decreased function; ibuprofen takes longer to be metabolized
CYP2C8	rs11572103	TT	A	0.006	CYP2C8*2 decreased function
CYP2C8	rs1058930	GG	C	0.04	CYP2C8*4 slightly decreased function
CYP2C8	rs7909236	TG	T	0.22	CYP2C8*1B Increased function

[Article: CYP2C9](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2C9	rs1799853	CC	T	0.1	CYP2C9*2 decreased function
CYP2C9	rs1057910	--	C	0.06	CYP2C9*3 decreased function
CYP2C9	rs2256871	AA	G	0.006	CYP2C9*9 decreased function
CYP2C9	rs9332131	--	D	0.0003	CYP2C9*6 decreased function
CYP2C9	rs28371685	CC	T	0.002	CYP2C9*11 decreased function

[Article: CYP2C19](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2C19	rs4244285	--	A	0.14	CYP2C19*2 decreased function; Plavix warning
CYP2C19	rs4986893	GG	A	0.007	CYP2C19*3 decreased function
CYP2C19	rs12248560	CT	T	0.22	CYP2C19*17 ultrafast metabolizer

[Article: CYP2D6](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2D6	rs3892097	CC	T	0.17	CYP2D6*4 decreased function
CYP2D6	rs5030655	--	D	0.001	CYP2D6*6 deletion; decreased function
CYP2D6	rs5030656	--	D	0.01	CYP2D6*9 decreased function
CYP2D6	rs1065852	GG	A	0.21	CYP2D6*10 decreased function
CYP2D6	rs28371725	--	T	0.09	CYP2D6*41 decreased function
CYP2D6	rs1135824	--	C	0.0007	CYP2D6*3 decreased function
CYP2D6	rs5030867	TT	G	0.0005	CYP2D6*7 decreased function
CYP2D6	rs28371706	GG	A	0.01	Possibly decreased function
CYP2D6	rs16947	--	A	0.32	CYP2D6*2 variant; possibly reduced function
CYP2D6	rs1135840	GC	G	0.57	G/G only: higher response rate to propranolol; adverse liver events with antituberculosis drugs[

[Article: CYP3A4](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP3A4	rs4987161	--	G	0.00006	CYP3A4*17 decreased function
CYP3A4	rs4986909	--	A	0.0001	CYP3A4*13 decreased function
CYP3A4	rs2740574	TT	C	0.11	CYP3A4*1B altered function
CYP3A4	rs4986910	--	G	0.006	CYP3A4*3 decreased function
CYP3A4	rs4986907	CC	T	0.002	CYP3A4*15A decreased function
CYP3A5	rs776746	CC	T	0.17	C/C: CYP3A5*3 non-functional –most common type in Caucasians, Asians; C/T: Partial CYP3A5 function; T/T: Functional CYP3A5
CYP3A5	rs28365083	--	T	0.005	CYP3A5*2 non-functional
CYP3A5	rs55817950	--	A	0.00002	CYP3A5*8 non-functional
CYP3A5	rs28383479	--	T	0.001	CYP3A5*9 non-functional
CYP3A5	rs41279854	--	G	0.00001	CYP3A5*10 non-functional
CYP3A5	rs56244447	--	T	0.0002	CYP3A5*3D non-functional
CYP3A4	rs28371759	AA	G	0.0005	CYP3A4*18, increased function; lower active vitamin D and lower BMD
CYP3A4	rs2242480	CC	T	0.15	TT: CYP3A4*36, decreased function; increased serum 1,25(OH)2D3
CYP3A4	rs35599367	GG	A	0.04	CYP3A4*22, reduced enzyme expression, slower metabolism of tacrolimus

[Article: CYP2E1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2E1	rs2031920	CC	T	0.02	Decreased risk of lung cancer and less likely to die from alcohol poisoning
CYP2E1	rs3813867	GG	C	0.009	Decreased risk of lung cancer and less likely to die from alcohol poisoning
CYP2E1	rs2515641	CT	T	0.13	lower risk of adverse reactions with Rifapentine

[Article: Allegra](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ABCB1	rs1128503	AG	G	0.58	GG: reduced drug efflux (less of the drug is moved out of the cell), thus may have a better response to fexofenadine (Allegra); A/G: intermediate efflux

[Article: CYP2J2](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2J2	rs890293	CC	A	0.07	40% lower expression of CYP2J2, increased risk of heart attack and high blood pressure, increased side effects from tacrolimus
CYP2J2	rs199717190	--	A	0	decreased clearance of terfenadine (drug metabolized by CYP2J2)
CYP2J2	rs1155002	CC	T	0.39	increased risk of COPD, increased risk of hypertension in women

Phase II Detoxification

[Article: SOD1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SOD1	rs1041740	CC	T	0.3	Decreased SOD1; increased risk of kidney problems with diabetes
SOD1	rs2070424	AA	G	0.08	Increased SOD1; protective against Alzheimer's and hearing loss
SOD1	i5000864	--	A	0	Rare mutations linked to ALS
SOD1	i5048521	--	C	0	Rare mutations linked to ALS
SOD1	rs10432782	--	G	0.07	Higher SOD1 enzyme activity, greater risk of noise-induced hearing loss
SOD1	rs121912436	--	A	0	Rare mutations linked to ALS
SOD1	rs121912431	--	C	0	Rare mutations linked to ALS

[Article: UGT family](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
UGT1A1*6	rs4148323	GG	A	0.006	Reduced activity; increased bilirubin
UGT1A1*60	rs4124874	TG	G	0.46	Reduced activity; increased bilirubin
UGT1A1*28	rs6742078	GT	T	0.32	TT only: increased bilirubin and gallstone risk
UGT1A1	rs35003977	--	G	0.0004	higher bilirubin
UGT1A6	rs17863783	GT	T	0.02	Increased UGT1A6
UGT1A8	rs6714486	TT	A	0.07	AA: increased UGT activity
UGT1A1	rs887829	CT	T	0.32	reduced UGT1A1 enzyme activity
UGT1A1	rs8330	GC	G	0.24	increased UGT1A1 enzyme activity
UGT2B15	rs1902023	AA	A	0.5	A/A: lower enzyme activity, possibly impairs BPA detoxification

[Article: Nrf2 Detoxification](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
NFE2L2	rs6726395	--	G	0.54	Increased Nrf2; greater lung volume in smokers
NFE2L2	rs13001694	AA	G	0.36	Increased Nrf2; reduced all cause mortality
NFE2L2	rs1806649	--	T	0.23	Increased Nrf2; Reduced risk of death in COPD
NFE2L2	rs6721961	GG	T	0.11	Diminished Nrf2 expression; increased lung cancer risk
NFE2L2	rs35652124	--	C	0.3	higher kidney disease risk[ref] higher oxidative stress damage in children with autism spectrum disorder

[Article: NATs](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
NAT1	rs4986782	GG	A	0.01	Slow acetylator
NAT1	rs15561	CA	A	0.28	Reduced function
NAT1	rs6586714	GG	A	0.13	Decreased risk of colon cancer from processed meat
NAT2	rs1495741	AG	A	0.74	AA: likely slow metabolizer; AG: likely intermediate metabolizer; GG: likely rapid metabolizer
NAT2	rs1208	GA	G	0.42	GG: likely slower metabolizer, increased risk of ALL

[Article: NQO1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
NQO1*3	rs1131341	--	A	0.03	Decreased function; Increased risk of cancers from benzene; involved in vit. K
NQO1	rs1800566	AG	A	0.2	Decreased function; Higher cancer risk due to benzene toxicity
NQO1	rs10517	GG	A	0.13	A/A: may need altered (lower) Warfarin dose

[Article: GSTs](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GSTM1	rs366631	--	A	0.78	AA: GSTM1 deletion
GSTA1	rs3957357	--	A	0.38	Low (AG) or non-functioning (AA) enzyme
GSTP1	rs1695	AA	G	0.32	Somewhat reduce enzyme function
GSTO1	rs4925	CC	A	0.3	Altered enzyme function; increased risk of PCOS

[Article: SULT Genes](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SULT1A1	rs1042028	--	T	0.31	SULT1A1*2, reduced enzyme function, impacts tamoxifen, breast cancer risk, lung cancer risk
SULT1A1	i6018900	--	T	0.31	SULT1A1*2, reduced enzyme function, impacts tamoxifen, breast cancer risk, lung cancer risk
SULT1A1	rs1801030	--	C	0.007	SULT1A1*3, moderately lower SULT1A1 activity
SULT1A1	rs1042157	--	A	0.32	lower SULT1A1 activity
SULT2A1	rs2547231	AA	C	0.15	lower relative risk of gallstones (likely lower sulfated bile acids)
SULT2A1	rs182420	TT	C	0.23	somewhat lower levels of DHEAS
SULT2A1	rs2637125	GG	A	0.13	lower levels of DHEAS
SULT2A1	rs2910397	CC	T	0.3	higher levels of DHEAS
SULT1C4	rs1402467	GG	G	0.26	lower sulfation and excretion of hesperidin
SULT1E1	rs3736599	CC	T	0.1	T/T: lower SULT1E1 activity on estrogen, higher risk of stroke; T/T or C/T: higher risk of endometrial cancer

Detoxification of Specific Toxicants

[Article: Phthalate metabolism](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GSTP1	rs1695	AA	G	0.32	Reduced function; slower elimination of phthalates
SOD2	rs5746136	CT	T	0.28	Higher phthalate levels; increased asthma risk
CYP2B6	rs3745274	TT	T	0.25	TT: decreased CYP2B6 enzyme needed for eliminating phthalates
GSTM1	rs366631	--	A	0.78	AA: deletion (null) GSTM1 gene; Women with phthalate exposure are at a 5-fold increased risk of fibroids.
HSPA1L	rs2227956	AA	G	0.15	G/G: decreased HSP70; insulin resistance not tied to phthalate exposure
CYP2C9	rs1799853	CC	T	0.1	decreased breakdown and clearance of phthalates
CYP2C8	rs11572080	CC	T	0.08	decreased breakdown and clearance of phthalates

Article: BPA Detoxification

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COX2	rs5277	--	C	0.88	C/C: increase in risk for liver dysfunction with BPA exposure
CAT	rs769217	CC	T	0.23	Increased risk of liver dysfunction in the elderly with BPA exposure
SOD2	rs4880	AA	A	0.51	Increased risk of liver dysfunction with BPA exposure
UGT2B15	rs1902023	AA	A	0.5	AA: decreased enzyme activity
UGT1A1	rs34983651	--	I	0	Decreased liver clearance of BPA
SULT1A1	i6018900	--	T	0.31	low activity of the enzyme
SULT1A1	rs9282861	--	T	0.31	low activity of the enzyme
ABCB4	rs2230028	CT	C	0.08	reduced ABCB4 activity, increased gallstone risk
CYP17A1	rs743572	GG	G	0.39	higher CYP17A1 function (possibly higher sex hormone levels); increased risk of prostate cancer, increased risk of PCOS, increased risk of acne vulgaris, and decreased risk of breast cancer in certain situations

Article: BChE

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
BChE	rs1803274	CT	T	0.19	K-variant; Decreased BChE; possible problems w/ organophosphates and nightshades
BChE	rs1799807	TT	C	0.01	A-variant; possibly delayed recovery from succinylcholine (anesthesia) (important)
BChE	rs28933389	GG	A	0.0008	F1-variant; possibly delayed recovery from succinylcholine (anesthesia) (important)
BChE	rs28933390	CC	A	0.005	F2-variant; possibly delayed recovery from succinylcholine (anesthesia) (important)
BChE	rs2668207	--	C	0.04	Minor decrease in BChE levels
BChE	rs1126680	--	T	0.07	Decreased BChE; Increased risk of hyperhidrosis combined with K-variant

Article: Tamiflu Response

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CES1	rs71647871	--	T	0.01	Decreased CES1 function; possibly poor response to Tamiflu
CES1	rs121912777	--	T	0.0008	Decreased CES1 function; possibly poor response to Tamiflu
ABCB1	rs1045642	GG	A	0.49	AA: reduced drug efflux from cell; increased risk adverse events from Tamiflu

Article: Opioid Receptors

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
OPRM1	rs1799971	GG	G	0.13	Reduced opioid receptor; decreased response to opioids and increased risk for dependence
OPRM1	rs10485057	--	G	0.09	Increased risk of alcohol use disorder
OPRM1	rs2281617	CC	T	0.14	Less euphoria with amphetamine
OPRM1	rs510769	CC	T	0.25	Less euphoria depending on dose of amphetamine
OPRD1	rs1042114	TT	G	0.12	disrupts the way the receptor works; increased risk of opiate addiction
OPRD1	rs569356	--	G	0.12	increased OPRD1 expression, increased risk of opioid addiction
OPRK1	rs702764	TT	C	0.15	decreased risk of impulse control disorder (Parkinson's patients), increased risk of heroin addiction when combined with rs1799971 G allele

Article: Neonicotinoid Pesticides

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CHRNA4	rs1044396	AA	A	0.49	Possibly less affected by neonicotinoids
CYP3A4	rs4987161	--	G	0.00006	Altered enzyme function; possibly more problems from neonicotinoids.
CYP3A4	rs4986909	--	A	0.0001	Altered enzyme function; possibly more problems from neonicotinoids.
CYP3A4	rs2740574	TT	C	0.11	Altered enzyme function; possibly more problems from neonicotinoids.
CYP3A4	rs4986910	--	G	0.006	Altered enzyme function; possibly more problems from neonicotinoids.
CYP3A4	rs4986907	CC	T	0.002	Altered enzyme function; possibly more problems from neonicotinoids.

Article: Glyphosate

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP1A1	rs1048943	TT	C	0.04	More likely to have low acetylcholinesterase with glyphosate exposure.

[Article: Organophosphate pesticides](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
BCHE	rs1803274	CT	T	0.19	Increased risk of Parkinson's from organophosphate exposure
PON1	rs662	TC	C	0.31	Increased risk of problems from organophosphate pesticides
Inter-gen	rs4242382	AA	A	0.1	Exposure to permethrin increases the risk of prostate cancer
CYP2B6	rs3745274	TT	T	0.25	TT and GT: decreased conversion of chlorpyrifos (organophosphate pesticide) into a toxic metabolite; GG: typical - higher formation of the toxic metabolite from chlorpyrifos and greater risk of toxicity in agricultural workers
CYP2B6	rs2279343	--	G	0.25	GG and AG: decreased conversion of chlorpyrifos (organophosphate pesticide) into a toxic metabolite; AA: typical - higher formation of the toxic metabolite from chlorpyrifos and greater risk of toxicity in agricultural workers

[Article: Arsenic](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
AS3MT	rs11191439	TT	C	0.1	An increased risk of skin lesions with arsenic exposure
AS3MT	rs3740393	--	C	0.08	CC: more arsenic excreted
GSTO1	rs4925	CC	A	0.3	Increased skin lesions and decreased arsenic excretion
MTHFR	rs1801133	GG	A	0.33	Increased skin lesions and decreased arsenic excretion

[Article: Mycotoxins](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
XPC	rs2228001	TG	G	0.4	Increased relative risk of liver cancer with aflatoxin B1 exposure
CYP1A2	rs12720461	CC	T	0.002	Decreased CYP1A2 enzyme activity, which may impact detoxification of aflatoxin B1
CYP1A2	rs72547517	--	A	0	Decreased CYP1A2 enzyme activity, which may impact detoxification of aflatoxin B1
CYP1A2	rs72547515	--	A	0.00004	Decreased CYP1A2 enzyme activity, which may impact detoxification of aflatoxin B1
CYP3A4	rs4987161	--	G	0.00006	CYP3A4*17, decreased function of enzyme; involved in phase I detoxification of aflatoxin G1
CYP3A4	rs4986909	--	A	0.0001	CYP3A4*13, decreased function of enzyme; involved in phase I detoxification of aflatoxin G1
CYP3A4	rs2740574	TT	C	0.11	CYP3A4*1B, decreased function of enzyme; involved in phase I detoxification of aflatoxin G1
CYP3A4	rs4986910	--	G	0.006	CYP3A4*3, decreased function of enzyme; involved in phase I detoxification of aflatoxin G1
CYP3A4	rs4986907	CC	T	0.002	CYP3A4*15A, decreased function of enzyme; involved in phase I detoxification of aflatoxin G1
GSTM1	rs366631	--	A	0.78	A/A: deletion (null) GSTM1 gene; increased risk of liver cancer with aflatoxin B1 exposure
GSTA1	rs3957357	--	A	0.38	GSTA1*B, low/ non-functioning enzyme; increased risk of kidney disease with ochratoxin A exposure
GSTP1	rs1695	AA	G	0.32	G/G: reduced function, increased risk of liver damage with aflatoxin B1 exposure
SLCO1B1	rs4149056	TT	C	0.14	Increased risk of liver damage with aflatoxin B1 exposure
ADRA1A	rs1048101	AG	G	0.48	G/G: This genotype may have fewer problems with reactions to patulin (theoretical); A/G and A/A: typical
GSDMB	rs7216389	TT	T	0.53	increased risk of childhood asthma; 3-fold increased risk of asthma with known mold exposure
IL13	rs20541	AA	A	0.2	higher IgE levels; increased risk of allergic bronchopulmonary aspergillosis (fungal infection of the lung); increased risk of allergies, COPD

Article: Multiple Chemical Sensitivities

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2C9	rs1799853	CC	T	0.1	T/T: CYP2C9*2 – poor metabolizer; increased susceptibility of MCS
CYP2C9	rs1057910	--	C	0.06	C/C: CYP2C9*3 – poor metabolizer; increased susceptibility to MCS
GSTP1	rs1695	AA	G	0.32	increased susceptibility to MCS
SOD2	rs5746136	CT	T	0.28	increased susceptibility to MCS
CAT	rs1001179	TC	T	0.2	increased susceptibility to MCS
GSTM1	rs366631	--	A	0.78	A/A: deletion (null) GSTM1 gene; increased relative risk of MCS
NOS3	rs2070744	TT	C	0.34	CC or CT: typical nitrate levels in MCS, no increased risk; TT: most common genotype, higher nitrate levels in MCS patients when exposed to triggers
ADORA2A	rs2298383	CC	C	0.44	lower relative risk of multiple chemical sensitivity (protective)
ADORA2A	rs5751876	TT	T	0.42	increased relative risk of multiple chemical sensitivity
TRPA1	rs920829	CC	T	0.12	increase airway sensitivity, increased heat pain sensitivity

Article: Mercury

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GSTM1	rs366631	--	A	0.78	AA: deletion (null) GSTM1 gene. Decreased mercury detoxification.
GSTP1	rs1695	AA	G	0.32	GG: reduced function; decreased mercury detoxification, higher mercury levels
GCLM	rs41303970	--	A	0.17	Higher blood mercury levels; Decreased/slower mercury detoxification
CPOX4	rs1131857	TT	G	0.14	Increased sensitivity to neurobehavioral effects of mercury

Article: Acetaminophen Liver Damage

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CD44	rs1467558	CT	T	0.16	TT: up to a 4-fold increase in risk of liver failure with chronic acetaminophen use; CT: elevated liver enzymes with higher doses of acetaminophen
CYP3A5	rs776746	CC	T	0.17	functioning CYP3A5, slightly increased relative risk of acetaminophen-induced liver toxicity
UGT1A9	rs3832043	--	I	0.49	reduced clearance of acetaminophen through glucuronidation
UGT1A1	rs8330	GC	G	0.24	increased UGT1A1 activity, faster metabolism of acetaminophen, lower risk of liver problems with acetaminophen
UGT2B15	rs1902023	AA	A	0.5	A/A: lower enzyme activity, lower acetaminophen glucuronide-to-acetaminophen concentration ratios, higher NAPQI

[Article: Lead Exposure](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ALAD	rs1800435	CG	G	0.08	possible higher blood lead levels when exposed to lead (not all studies agree); more susceptible to kidney problems from lead exposure
HFE	rs1800562	--	A	0.05	C282Y mutation; increased QT interval with higher blood lead; higher levels of lead in bones
HFE	rs1799945	CC	G	0.13	H63D variant; possibility of increased QT interval with higher blood lead; possibly higher levels of lead in bones
SLC11A2	rs224589	--	T	0.24	higher blood lead levels
GPX1	rs1050450	GG	A	0.32	increased risk of brain tumors with higher lead levels
GSTM1	rs366631	--	A	0.78	A/A only: GSTM1 null, common in many population groups, increased inflammation from higher blood lead levels
TNF	rs1800629	GG	A	0.15	higher TNF-alpha (inflammatory cytokine); more inflammation in metal workers exposed to lead
TNF	rs1799964	TC	C	0.21	C/C: usually higher TNF-alpha (inflammatory cytokine); more inflammation in metal workers exposed to lead

[Article: DPYD and fluorouracil](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
DPYD	rs3918290	--	T	0.004	DPYD*2A mutation, reduced activity, may react negatively to 5-fluorouracil or capecitabine
DPYD	rs55886062	--	C	0.0006	reduced activity, may react negatively to 5-fluorouracil or capecitabine
DPYD	rs1801160	CC	T	0.05	reduced activity, may react negatively to 5-fluorouracil or capecitabine
DPYD	rs67376798	TT	A	0.005	moderately reduced activity, may react negatively to 5-fluorouracil or capecitabine
DPYD	rs75017182	--	C	0.004	moderately reduced activity, may react negatively to 5-fluorouracil or capecitabine

Article: Glutathione

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GSS	rs121909307	--	T	0.00008	glutathione synthase deficiency
GSR	rs8190955	GG	A	0.008	significantly decreased GSR (rare); less glutathione to use as an antioxidant, higher oxidative stress levels
GSR	rs2978663	TT	C	0.38	higher bone mineral density; likely higher glutathione levels
GGT1	rs5751901	CC	C	0.37	increased risk of pancreatitis, especially in smokers, increased risk of pancreatitis with ERCP
GGT1	rs4820599	AA	G	0.32	increased risk of pancreatitis
GCGL	rs17883901	GA	A	0.07	lower activity, increased relative risk of myocardial infarction
GCGL	rs2301022	--	T	0.32	decreased risk of stroke
GCGL	rs17883718	GG	A	0.002	rare mutation, significantly decreased enzyme function
GPX1	rs1050450	GG	A	0.32	Lower GPX 1 enzyme activity, increased risk of breast cancer; higher selenium plasma levels, linked to Keshan disease in people with low selenium levels
GPX1	rs1800668	--	A	0.28	increased oxidative stress after exercise (elite athletes)[ref] increased risk of schizophrenia, increased risk of cataracts in diabetic patients
GPX4	rs713041	CC	T	0.44	needs sufficient selenium intake; increased risk of stroke, colon cancer; decreased risk of pre-eclampsia and endometriosis, increased risk of pancreatitis in smokers
GSTM1	rs366631	--	A	0.78	deletion (null) GSTM1 gene. 2x increased risk of breast cancer, increased susceptibility to lung cancer in smokers, increased risk of prostate cancer in Caucasians
GSTM5	rs3754446	AC	C	0.33	increased oxidative stress, lower survival rates in leukemia patients
GSTA1	rs3957357	--	A	0.38	GSTA1*B, low/ non-functioning enzyme; increased risk of asthma, allergies; interaction with cruciferous vegetables and cancer risk
GSTP1	rs1695	AA	G	0.32	reduced function, increased risk of breast cancer; increased risk of prostate cancer; increased risk of nasal polyps
GSTO1	rs4925	CC	A	0.3	lower mortality in renal cell carcinoma (males); increased risk of PCO; increased problems detoxifying arsenic

[Article: pyrethroid metabolism](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2C19	rs4244285	--	A	0.14	non-functioning (CYP2C19*2)
CYP2C19	rs4986893	GG	A	0.007	poor metabolizer (CYP2C19*3), increased risk of lung cancer
CYP2C9	rs1799853	CC	T	0.1	CYP2C9*2 – poor metabolizer; 40% reduction in Warfarin metabolism
CYP2C9	rs1057910	--	C	0.06	CYP2C9*3 – poor metabolizer; 80% reduction in Warfarin metabolism
CYP2C9	rs2256871	AA	G	0.006	CYP2C9*9 – poor metabolizer
CYP2C9	rs28371685	CC	T	0.002	CYP2C9*11 – poor metabolizer
CES1	rs121912777	--	T	0.0008	impaired enzyme function, carboxylesterase 1 deficiency
HLA-DRA	rs3129882	GA	G	0.44	GG: Increased HLA-DRA function, increased risk of Parkinson's with pyrethroid exposure (~2.5x)
Chr 8q24	rs4242382	AA	A	0.1	Exposure to permethrin increases the risk of prostate cancer
CYP2C9	rs9332131	--	D	0.0003	*deletion* or DD: CYP2C9*6 – poor metabolizer

[Article: ABCB1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ABCB1	rs1128503	AG	A	0.41	A/A: greater efflux, transport out of the cell; A/G: intermediate transport
ABCB1	rs4148740	AA	G	0.13	G/G: better response to amitriptyline, citalopram, paroxetine, or venlafaxine in depression patients; lower risk of chronic Lyme after antibiotic treatment, likely less excretion (lower ABCB1 function)
ABCB1	rs2032583	AA	G	0.13	higher plasma drug levels, more sleep-related side effects with antidepressants; increased risk of adverse events from citalopram, fluvoxamine, paroxetine, sertraline, venlafaxine, likely less excretion (lower ABCB1 function)
ABCB1	rs2229109	CC	T	0.03	reduced cellular accumulation of aldosterone, dexamethasone, and cortisol; reduced effectiveness of topical steroids; increased ABCB1 transport when exposed to progesterone, estrogen, likely more excretion (higher ABCB1 function)
ABCB1	rs2214102	CC	T	0.06	no increased risk of breast cancer with estrogen/progesterone HRT, likely more excretion (higher ABCB1 function)
ABCB1	rs3842	CT	C	0.14	Increased risk of lung cancer in smokers, increased risk of bleeding with direct oral anticoagulants, likely less excretion of toxins/drug (lower ABCB1 function)

[Article: PFAS](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP1A1	rs1048943	TT	C	0.043	increased risk of breast cancer in women with higher PFAS levels
COMT	rs4680	AG	A	0.49	A/A (Met/Met) lower COMT activity; increased risk of breast cancer in women with higher PFAS levels
MTR	rs1805087	AA	G	0.19	greater birth weight reductions from maternal high PFAS levels
PPARGC1A	rs8192678	CC	T	0.32	decreased body fat; decreased palmitic, palmitoleic, and oleic acid levels in pregnant women with higher PFAS levels
FTO	rs9939609	TT	A	0.39	increased FTO expression, increased insulin resistance in adulthood due to PFAS exposure.
GSTM1	rs366631	--	A	0.79	A/A: deletion (null) GSTM1 gene. 2x increased risk of atopic dermatitis (eczema) with higher than median PFAS exposure
CYP17A1	rs743572	GG	G	0.39	higher CYP17A1 function (possibly higher sex hormone levels); higher androgen levels in female babies born to mothers with higher PFAS levels; lower risk of breast cancer with high PFAS exposure